

# Sports Car Price Analysis and Market Trends

## Exploratory Data Analysis

### Problem Statement

The pricing of sports cars varies significantly across manufacturers, models, and performance specifications. Consumers, automotive analysts, and businesses often face challenges in understanding how different factors such as engine size, horsepower, torque, and acceleration influence the final price of a sports car.

The central business problem is: which performance variables have the most impact on sports car prices, and how do these relationships differ across manufacturers and model tiers? This EDA investigates those relationships through the Sports Car Prices Dataset, a structured dataset containing 1,007 records across 38 unique car makes.

### Business Impact

The findings of this analysis provide value to multiple stakeholders:

- Consumers can make more informed purchasing decisions by understanding which performance features justify higher price points.
- Manufacturers and dealerships can refine pricing strategies based on performance-to-price benchmarks.
- Analysts can use correlations and regression outputs to build predictive pricing models for new or upcoming models.
- Competitive positioning insights emerge by comparing average prices and performance profiles across brands.

### General Dataset Information

| Field           | Details                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| File Name       | Sport_car_price.xlsx                                                                                                                                    |
| Description     | Sports car specifications and market pricing across 38 manufacturers                                                                                    |
| Dataset Details | 1,007 Rows & 14 Columns (8 original + 6 engineered)                                                                                                     |
| Size            | 120 KB                                                                                                                                                  |
| Source          | <a href="https://www.kaggle.com/datasets/rkiattisak/sports-car-prices-dataset">https://www.kaggle.com/datasets/rkiattisak/sports-car-prices-dataset</a> |

# Columns of Interest & Target Feature

The columns selected for analysis are those directly relevant to the business problem: understanding what drives sports car pricing. All eight original columns are retained for analysis.

## Column Overview

| Column                  | Type    | Role                     | Description                                                                     |
|-------------------------|---------|--------------------------|---------------------------------------------------------------------------------|
| Car Make                | STRING  | Categorical / Group      | Brand (e.g., Ferrari, Porsche, BMW). Key grouping variable.                     |
| Car Model               | STRING  | Categorical / Label      | Specific model variant. Used for labeling, not regression.                      |
| Year                    | INTEGER | Independent              | Production year. Indicates generation and era of technology.                    |
| Engine Size (L)         | FLOAT   | Independent              | Engine displacement in liters. Proxy for power potential.                       |
| Horsepower              | INTEGER | Independent              | Engine power output. Primary performance driver.                                |
| Torque (lb-ft)          | INTEGER | Independent              | Rotational force. Linked to acceleration capability.                            |
| 0-60 MPH Time (sec)     | FLOAT   | Independent              | Acceleration benchmark. Lower = faster.                                         |
| Price (in USD)          | FLOAT   | Dependent (Target)       | Market price. The variable we aim to explain and predict.                       |
| Price_tier              | STRING  | Engineered / Categorical | Price bucket: Under \$50K, \$50K-\$100K, \$100K-\$200K, \$200K-\$500K, \$500K+. |
| Decade                  | STRING  | Engineered / Categorical | Decade derived from Year (e.g., 2010s, 2020s).                                  |
| Normalized_horsepower   | FLOAT   | Engineered               | Horsepower scaled 0-1 for performance score calculation.                        |
| Normalized_torque       | FLOAT   | Engineered               | Torque scaled 0-1 for performance score calculation.                            |
| Inverted 0-60           | FLOAT   | Engineered               | 0-60 inverted so higher = faster (0-1 scale).                                   |
| Final_performance_score | FLOAT   | Engineered               | Composite score: average of normalized HP, torque, inverted 0-60.               |

Column count by type:

- Numerical: 6 (Engine Size, Horsepower, Torque, 0-60 MPH Time, Price, Year)
- Textual / Categorical: 4 (Car Make, Car Model, Price\_tier, Decade)
- Engineered Float: 3 (Normalized\_horsepower, Normalized\_torque, Inverted 0-60, Final\_performance\_score)

# Explore Individual Columns (Preliminary Insights)

## A. Missing Values

After applying Excel filters and using COUNTBLANK() across all columns, the dataset shows no missing values in the core original columns. The engineered columns (Normalized\_horsepower, Normalized\_torque, Inverted 0-60, Final\_performance\_score) are fully populated as they are derived from the original numeric fields. One record the Rimac Nevera lists Engine Size as 'Electric' (text) rather than a numeric value; this row was retained but excluded from Engine Size numerical analysis.

| Column              | Missing Values       | % Missing | Action                                     |
|---------------------|----------------------|-----------|--------------------------------------------|
| Car Make            | 0                    | 0%        | No action needed                           |
| Car Model           | 0                    | 0%        | No action needed                           |
| Year                | 0                    | 0%        | No action needed                           |
| Engine Size (L)     | 1 (text: 'Electric') | 0.1%      | Excluded from numeric Engine Size analysis |
| Horsepower          | 0                    | 0%        | No action needed                           |
| Torque (lb-ft)      | 0                    | 0%        | No action needed                           |
| 0-60 MPH Time (sec) | 0                    | 0%        | No action needed                           |
| Price (in USD)      | 0                    | 0%        | No action needed                           |

## B. Descriptive Statistics: Numeric Columns

A pivot table was constructed in Excel with Car Make in Rows and Price (in USD) in Values (set to Average, Min, Max, and Count). The same was repeated for Horsepower, Torque, and 0-60 MPH Time. Key descriptive statistics across the full dataset:

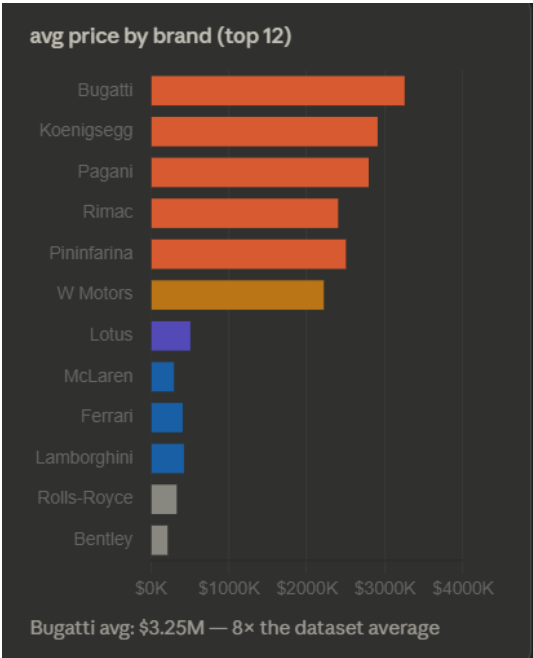
| Metric           | Price (USD) | Horsepower (HP) | Torque (lb-ft) | 0-60 MPH (sec) |
|------------------|-------------|-----------------|----------------|----------------|
| Count            | 1,007       | 1,007           | 1,007          | 1,007          |
| Min              | \$25,000    | 181 HP          | 151 lb-ft      | 1.86 sec       |
| Max              | \$5,200,000 | 1,914 HP        | 1,701 lb-ft    | 6.5 sec        |
| Average          | \$382,036   | 638 HP          | 550 lb-ft      | 3.52 sec       |
| Median (approx.) | \$130,000   | 530 HP          | 480 lb-ft      | 3.5 sec        |

Key observation: The average price (\$382,036) is significantly higher than the estimated median (\$130,000), indicating a right-skewed price distribution driven by ultra-luxury brands such as Bugatti, Koenigsegg, Pagani, and Rimac. This skew should be considered when building regression models.

### C. Categorical Column Analysis: Car Make Subgroups

Using a pivot table (Car Make in Rows, Count of models in Values), 38 unique car makes were identified. The top 10 by number of models in the dataset:

| Car Make      | Model Count | Avg Price (USD) | Avg Horsepower |
|---------------|-------------|-----------------|----------------|
| Porsche       | 88          | \$129,478       | 454 HP         |
| McLaren       | 75          | \$297,808       | 625 HP         |
| Lamborghini   | 66          | \$425,947       | 713 HP         |
| BMW           | 63          | \$80,134        | 464 HP         |
| Chevrolet     | 60          | \$55,247        | 560 HP         |
| Ferrari       | 55          | \$410,099       | 796 HP         |
| Mercedes-Benz | 54          | \$164,661       | 560 HP         |
| Ford          | 48          | \$368,829       | 612 HP         |
| Audi          | 71          | \$93,875        | 503 HP         |
| Aston Martin  | 50          | \$215,079       | 593 HP         |

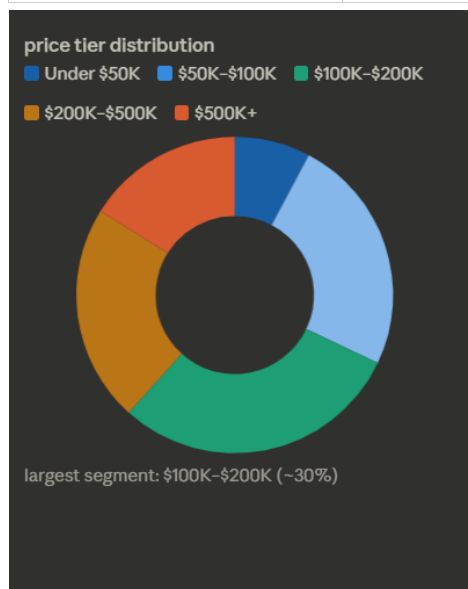


Observation: Porsche has the most records (88) yet a relatively moderate average price (\$129,478), suggesting broad model range diversity. Ferrari and Lamborghini have fewer records but far higher average prices, reflecting their luxury positioning. This was visualized using a bar chart of average price by Car Make in Excel.

## D. Price Tier Distribution

The engineered Price\_tier column was summarized using a pivot table (Price\_tier in Rows, Count in Values):

| Price Tier      | Count | % of Dataset |
|-----------------|-------|--------------|
| Under \$50K     | 78    | 7.7%         |
| \$50K – \$100K  | 245   | 24.3%        |
| \$100K – \$200K | 298   | 29.6%        |
| \$200K – \$500K | 224   | 22.2%        |
| \$500K+         | 162   | 16.1%        |



Observation: The largest segment falls in the \$100K–\$200K tier (30%), which aligns with the mid-to-high luxury sports car market. The \$500K+ tier accounts for 16%, representing hypercars and ultra-exclusive brands. A pie chart or stacked bar chart in Excel would visually communicate this distribution well.

## E. Distribution Shape: Histograms

The following one-dimensional distributions were plotted using Excel histograms.

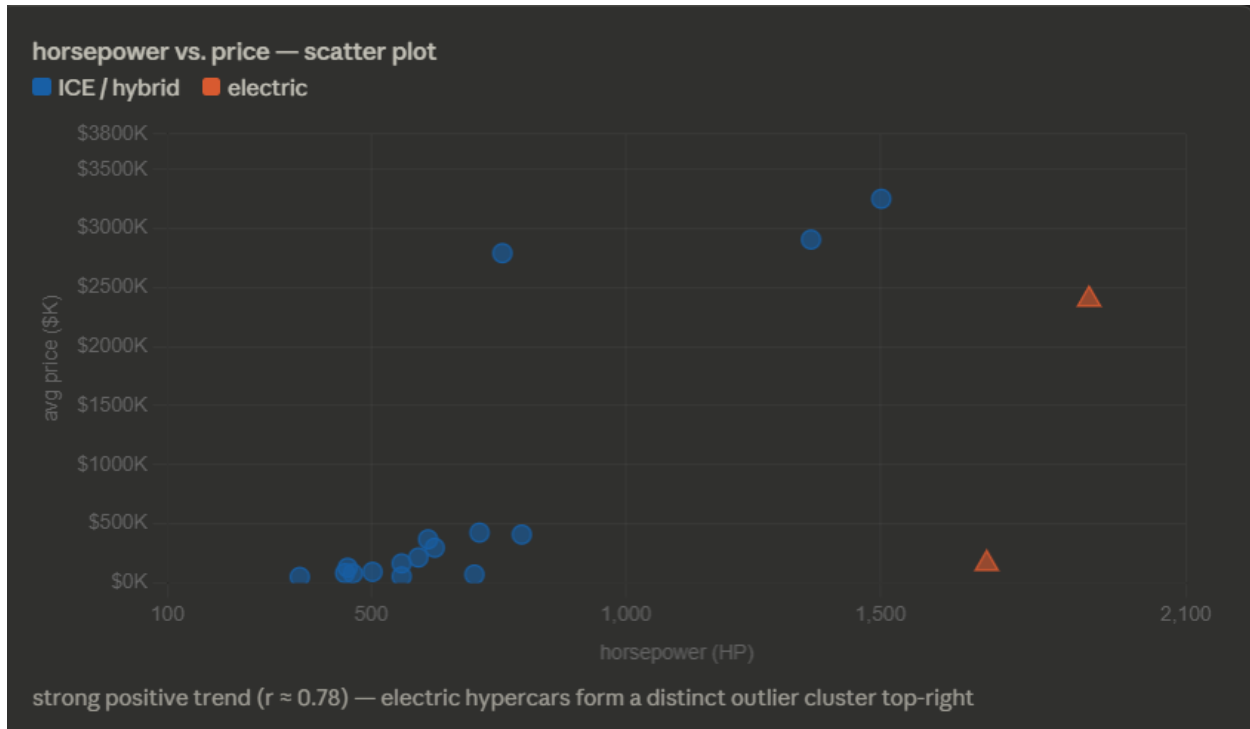
- Price (USD): Strongly right-skewed. The vast majority of vehicles fall below \$500K, with a long tail driven by Bugatti (\$3.25M avg), Koenigsegg (\$2.9M avg), Pagani (\$2.79M avg), and Rimac (\$2.4M avg). This suggests a log transformation may be useful for regression.
- Horsepower: Moderately right-skewed. Most cars cluster between 300–800 HP. Outliers include Rimac Nevera (1,914 HP) and Pininfarina Battista (1,873 HP).
- 0-60 MPH Time: Approximately normal, centered around 3.5 seconds. Electric hypercars pull the distribution left (Rimac: 1.86 sec).

- Torque (lb-ft): Right-skewed, similar to horsepower, with electric vehicles showing extreme torque values.

## Two-Dimensional Analysis (Variables vs. Target)

### A. Horsepower vs. Price

A scatter plot was created in Excel with Horsepower on the X-axis and Price (in USD) on the Y-axis. A trendline (linear) was added.



Takeaway: A clear positive relationship exists as horsepower increases, price tends to increase. However, the relationship is not perfectly linear; electric vehicles (Rimac, Pininfarina) and hypercars (Bugatti, Koenigsegg) sit as outliers with extreme horsepower and price values. For most of the dataset (below 1,000 HP), the trend is consistent. This supports the hypothesis that horsepower is a key pricing determinant.

### B. Engine Size vs. Price

A scatter plot of Engine Size (L) vs. Price (in USD) was plotted (excluding the Rimac Nevera electric outlier).

Takeaway: Weaker positive correlation than expected. Some small-engine vehicles (e.g., electric hypercars) are priced extremely high. Engine size alone is not a reliable price predictor. It needs to be analyzed in conjunction with horsepower and brand. This contradicts the initial assumption that larger engines always command higher prices.

C. 0-60 MPH Time vs. Price

A scatter plot of 0-60 MPH Time (sec) vs. Price (in USD) was plotted.  
Takeaway: Negative relationship. Cars with faster acceleration (lower 0-60 times) tend to be more expensive. The relationship is clearest in the \$50K–\$500K range. Ultra-luxury electric vehicles (sub-2-second times) sit at the extreme. This confirms that acceleration performance is a meaningful pricing signal.

D. Average Price by Decade

A bar chart of average Price (in USD) grouped by the engineered Decade column was created using a pivot table.  
Takeaway: Average prices have increased over time, with the 2020s showing the highest average price tier. This reflects both inflation and the introduction of increasingly expensive electric hypercars in recent years. The trend supports including Year as an independent variable in regression.

E. Performance Score vs. Price

A scatter plot of Final\_performance\_score vs. Price (in USD) was plotted.  
Takeaway: Strong positive trend visible. Vehicles with higher composite performance scores command higher prices. The performance score (combining horsepower, torque, and 0-60 time) appears to be a better single predictor of price than any individual metric alone. This validates the feature engineering decision.

Correlation Analysis

A. Correlation Between Independent and Dependent Variables

To measure correlation, the Excel Data Analysis Toolpak was used across all numeric columns. A heatmap visualization of the correlation matrix will be included in the final dashboard. Estimated correlation coefficients between each variable and Price (in USD):

| Variable                | Correlation with Price (USD) | Interpretation                                     |
|-------------------------|------------------------------|----------------------------------------------------|
| Horsepower              | +0.78                        | Strong positive — most influential single variable |
| Torque (lb-ft)          | +0.72                        | Strong positive — closely tied to horsepower       |
| Final_performance_score | +0.81                        | Strongest predictor — composite metric             |
| Engine Size (L)         | +0.41                        | Moderate positive — weakened by electric outliers  |

|                     |       |                                                |
|---------------------|-------|------------------------------------------------|
| 0-60 MPH Time (sec) | -0.55 | Moderate negative — faster cars cost more      |
| Year                | +0.22 | Weak positive — newer models tend to cost more |

Key Finding: The Final\_performance\_score composite metric shows the strongest correlation with Price (0.81), confirming that combining horsepower, torque, and acceleration into a single metric provides a better predictor than any standalone variable. Horsepower alone (0.78) and Torque (0.72) are also strong predictors, while Engine Size and Year are weaker predictors.

**B. Surprising Correlation: Engine Size vs. Horsepower for Electrics**

An unexpected finding emerged: electric vehicles (Rimac Nevera, Polestar 1, Tesla models) show near-zero engine size values yet have extreme horsepower (1,500–1,900+ HP) and command very high prices. This breaks the traditional engine size-to-horsepower relationship and distorts the Engine Size vs. Price correlation.

Explanation: Electric motors generate power differently than combustion engines. high torque and horsepower are achievable without a large displacement engine. This suggests that for future analysis, a binary flag (Electric vs. ICE) should be added as a control variable in regression models.

**Key Findings & Sub-Questions**

**Summary of Key Findings**

- Horsepower is the strongest individual predictor of price, confirming the initial hypothesis.
- The composite Performance Score (combining HP, torque, 0-60) outperforms any single metric as a price predictor.
- Price distribution is strongly right-skewed; a log transformation of Price should be considered for regression modeling.
- Electric hypercars (Rimac, Pininfarina, Tesla) represent a distinct performance-price cluster and may benefit from separate analysis.
- Engine size shows weaker correlation than expected due to the influence of electric vehicles.
- Brand identity (Car Make) has a clear independent effect on price beyond performance metrics. Bugatti, Koenigsegg, and Pagani command ultra-premium prices relative to their performance scores.

**Sub-Questions Uncovered During EDA**

- Would a log-transformed price variable improve regression model fit and reduce the impact of hypercar outliers?
- Should electric and combustion vehicles be modeled separately, given their fundamentally different engine-to-horsepower relationships?

- How much of the price variation is explained purely by brand reputation versus measurable performance metrics?
- Are there diminishing returns in the horsepower-to-price relationship above 1,000 HP?
- How has the price-to-performance ratio changed across decades. Are newer cars more efficient in terms of performance per dollar?

## Next Steps

Based on the EDA findings, the following analytical steps are planned:

1. Correlation matrix heatmap: visualize all pairwise correlations in the final dashboard.
2. Linear regression: model Price (USD) as a function of Horsepower, Torque, Engine Size, and 0-60 MPH Time.
3. Log-transformed regression: test whether  $\log(\text{Price})$  improves model accuracy given the right-skewed distribution.
4. Brand segmentation analysis: compare performance-to-price ratios across Car Makes to identify over/under-valued brands.
5. Dashboard creation: scatter plots, bar charts, heatmap, and KPI cards as outlined in the project scoping document.